Lab Report: Text classification with neural networks

1 Introduction

In this report we compare several models on standard benchmarks. We trained each model for 20 epochs with a batch size of 128. The dataset was split into 80 percent training data and the rest for testing.

2 Results

Figure 1 shows the f1 score by dataset.

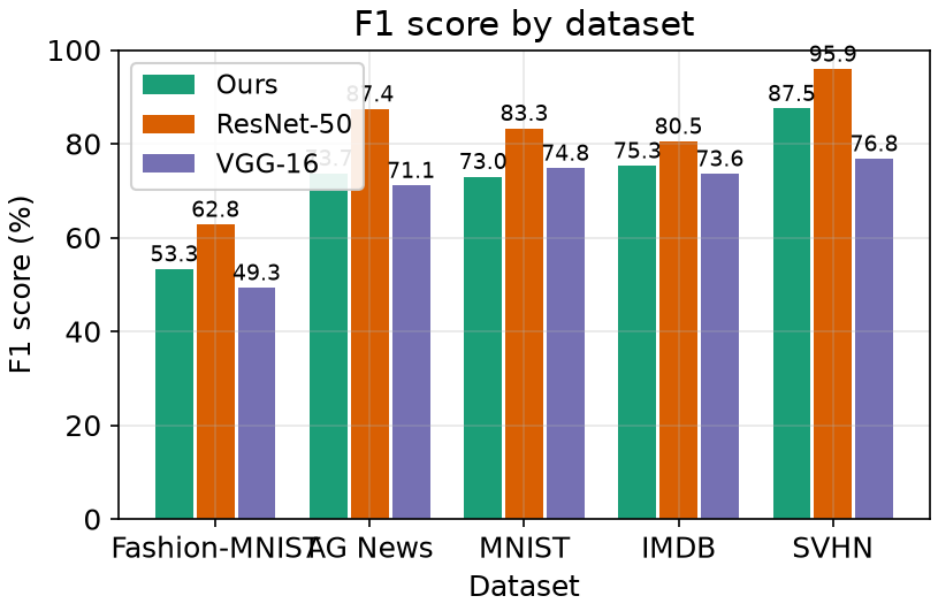


Figure 1: F1 score by dataset for the models in this experiment.

ResNet-50 achieves 145.9% F1 score on SVHN. ResNet-50 outperforms VGG-16 by 7 percentage points on IMDB. As shown in Figure 1, Ours reaches 80.5% F1 score on IMDB.

Figure 2 shows the test accuracy of CNN by dataset.

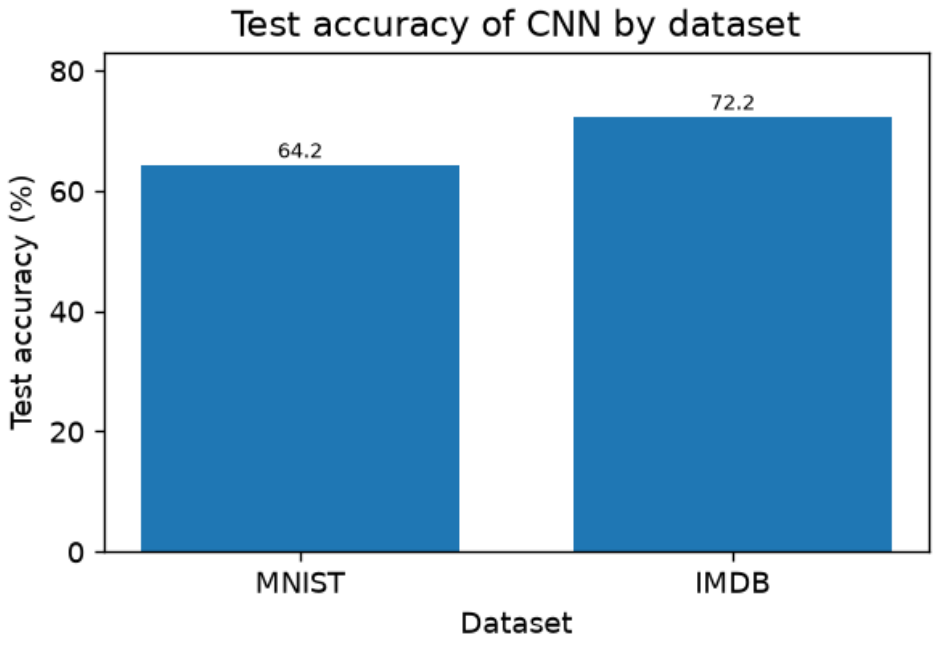


Figure 2: Test accuracy of CNN by dataset for the models in this experiment.

CNN reaches 64% test accuracy on MNIST (Figure 2). CNN scores 8.0 points higher on IMDB than on

MNIST. On IMDB, CNN obtains a test accuracy of 30.7% (Figure 2).

Figure 3 shows the accuracy during training.

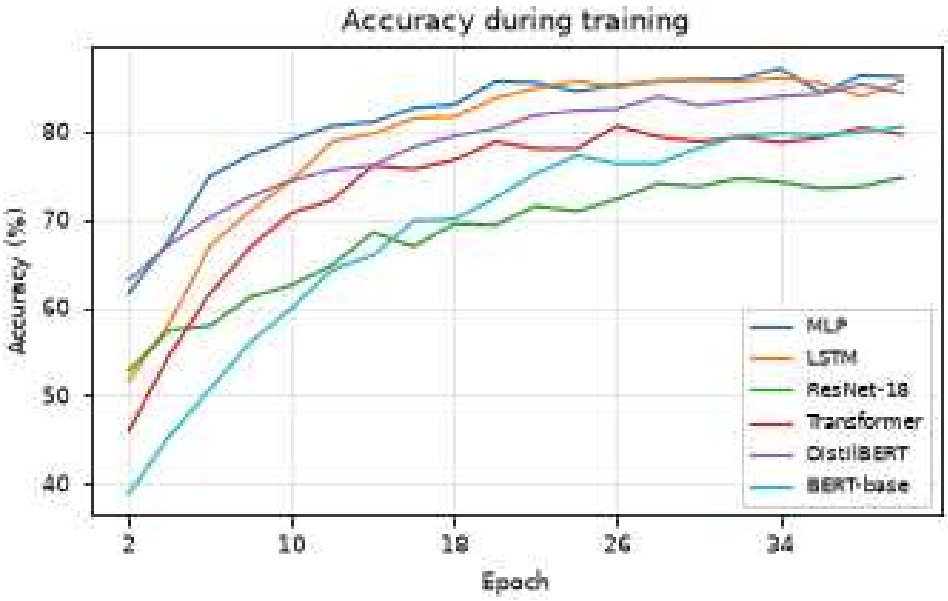


Figure 3: Accuracy during training for the models in this experiment.

From epoch 6 to epoch 8, the accuracy of LSTM improves by a relative 4.4%. From epoch 6 to epoch 28, the accuracy of BERT-base improves by a relative 50%. As shown in Figure 3, Transformer reaches 57.9% accuracy at epoch 6.

3 Discussion

Overall, the results are in line with what we expected from the literature.